

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI, KARNATAKA — 590018



REPORT ON

INTERNSHIP (22UAI803I)

AT

SuprMentr

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD

OF THE DEGREE OF

BACHELOR OF ENGINEERING

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

SUBMITTED BY

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USN: 2BA22AI053

INTERNAL GUIDE
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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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BASAVESHWAR ENGINEERING COLLEGE, S. NIJALINGAPPA VIDYANAGAR

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**BASAVESHWAR ENGINEERING
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ACADEMIC YEAR 2025-26



CERTIFICATE

This is to Certify that “Internship Report” submitted by **MR. PRAVEENKUMAR PATTAR (2BA23AI403)** is Bonafide student of **Basaveshwar Engineering College, Bagalkote**, in partial fulfillment for the award of **Bachelor of Engineering in Artificial Intelligence and Machine Learning** during the year **2025-2026**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library.

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1.

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DECLARATION

I, **PRAVEENKUMAR PATTAR (2BA22AI053)** student of 8th semester Bachelor of Engineering in Artificial Intelligence and Machine Learning, Basaveshwar Engineering College, Bagalkote, hereby declare that the Internship work entitled ” **Web App Security Testing Tool at SuprMentr**” submitted to the Visvesvaraya Technological University during the academic year 2025-2026, is a record of an original work done by me under the guidance of **Prof. Ashok S Patil**, and **Dr. Bharati M Reshmi**, Head of the department, Department of Artificial Intelligence and Machine Learning, Basaveshwar Engineering College, Bagalkote. This Internship work is submitted in fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Artificial Intelligence and Machine Learning. The result embodied in this is have not been submitted to any other University or institute for the award of any degree.

Place: BAGALKOTE

Date: 04-06-2026

PRAVEENKUMAR PATTAR

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CERTIFICATE BY THE INDUSTRY/COMPANY/ORGANIZATION



Certificate ID: SM26VCSEH0087

CERTIFICATE OF INTERNSHIP

This is to certify that

Praveenkumar Pattar

from

Basaveshwar Engineering College Bagalkot

has successfully completed a **90 Days** Internship at
SuprMentr Technologies Pvt Ltd in partnership with **VTU, Belagavi**
from **February 1st 2026 - May 1st 2026**

During the internship, the student has demonstrated exceptional
commitment and performance in the domain of

Cyber Security and Ethical Hacking

and developed a project titled

Web App Security Testing Tool

The internship culminated in the successful submission of a
comprehensive capstone project, showcasing technical proficiency,
creativity, and problem-solving abilities. We commend the student's
outstanding contributions and wish continued success in future
endeavours.

Supreeth Y S

Supreeth Y S
Director
SuprMentr Technologies



Yashas

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ACKNOWLEDGEMENT

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We express our sincere gratitude to our Principal, **Dr. Basavaraj R. Hiremath**, Basaveshwar Engineering College, for providing a supportive academic environment and the necessary facilities for the successful completion of the internship.

We would like to express our heartfelt thanks to our Head of the Department, **Dr. Bharati M. Reshmi**, Department of Artificial Intelligence and Machine Learning Engineering, for her encouragement and valuable support throughout the internship period.

We are deeply grateful to our Internship Internal Guide, **Prof. Ashok S. Patil**, for his continuous guidance, motivation, and valuable suggestions which greatly helped us during the internship work.

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ABSTRACT

The project titled “**Web App Security Testing Tool – Automate Testing for XSS and SQL Injection Vulnerabilities**” is a web-based cybersecurity application developed to automate the process of identifying common security vulnerabilities and security misconfigurations in web applications. With the increasing number of cyber threats targeting websites and online services, there is a growing need for efficient tools that can assist developers and security professionals in assessing the security posture of web applications.

The system was developed using **Python** and the **Flask** web framework, with a modern and interactive frontend designed using **HTML, CSS, and JavaScript**. The application allows users to enter a target website URL and perform automated security assessments. The tool analyzes important security aspects including **Security Header Evaluation, SQL Injection Detection, Cross-Site Scripting (XSS) Testing, Technology Fingerprinting, Open Port Analysis, Vulnerability Detection, Risk Assessment, and Security Recommendation Generation**.

The internship project provided practical exposure to **Cyber Security, Ethical Hacking, Vulnerability Assessment, Secure Software Development, Web Technologies, Network Security Concepts, and Python-Based Application Development**. The developed system demonstrates how automated security testing can assist in identifying potential security weaknesses and improving the overall security posture of web applications.

Future enhancements may include the integration of **Artificial Intelligence for intelligent threat analysis, advanced penetration testing modules, automated report generation, vulnerability trend analysis, and support for additional web security testing techniques**, making the system more effective and suitable for real-world cybersecurity applications.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

Abbreviation	Description
XSS	Cross-Site Scripting
SQLi	SQL Injection
OWASP	Open Web Application Security Project
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
CSP	Content Security Policy
API	Application Programming Interface
UI	User Interface
UX	User Experience
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript

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CHAPTER 1

COMPANY PROFILE

Company Name: SuprMentr Technologies Private Limited

Year of Establishment: 2023

Address: 48, Church St, Haridevpur, Shanthala Nagar, Ashok Nagar, Bengaluru-560001

Type of Company: Private Sector Organization

Mission Statement:

The mission of SuprMentr Technologies Private Limited is to bridge the gap between academic education and industry requirements by providing career mentorship, industry-oriented training, internship opportunities, and practical exposure to emerging technologies. The organization aims to empower students with technical, professional, and problem-solving skills required to excel in the modern workforce.

Clients:

Engineering Students, Educational Institutions, Universities and Colleges, Fresh Graduates, Technology Learners, Industry Professionals seeking Upskilling Opportunities.

Domains of Company Expertise:

- Career Mentorship and Guidance
- Internship and Industry Training Programs
- Cyber Security and Ethical Hacking
- Cloud Computing
- Professional Skill Development
- Placement Preparation and Career
- Web Development
- Software Development Training

SuprMentr Technologies Private Limited is a career-tech and ed-tech organization that focuses on providing industry-oriented training, internship opportunities, mentorship programs, and practical learning experiences for students. The organization helps students bridge the gap between academic education and industry requirements by offering hands-on exposure to emerging technologies, professional skill development, and real-world project implementation.

During the internship period, the organization provided practical exposure in the field of Cyber Security and Ethical Hacking, including web application security, vulnerability assessment, ethical hacking methodologies, cybersecurity best practices, and secure software development. Students were encouraged to work on innovative projects that address real-world cybersecurity challenges and promote secure digital environments.

As part of the internship, the project titled “Web App Security Testing Tool – Automate Testing for XSS and SQL Injection Vulnerabilities” was developed in the domain of Cyber Security and Ethical Hacking. The project focused on automating web application security assessment through SQL Injection detection, Cross-Site Scripting (XSS) testing, security header analysis, technology fingerprinting, port scanning, attack surface evaluation, and risk assessment. The system was designed to identify common security weaknesses in web applications and provide recommendations for improving security posture.

CHAPTER 2

INTERNSHIP DESCRIPTION

An internship is a professional learning experience that provides meaningful and practical exposure related to a student's academic background and career interests. It serves as an important platform for bridging the gap between theoretical concepts and real-world applications. Internships help students develop technical knowledge, problem-solving abilities, teamwork, communication skills, and professional discipline required in industry environments.

For students in the field of Cyber Security and Ethical Hacking, internships play a significant role in gaining practical experience in web application security, vulnerability assessment, ethical hacking methodologies, risk analysis, and secure software development practices. Such opportunities help students understand security testing methodologies, vulnerability detection techniques, penetration testing concepts, and implementation strategies used in real-world cybersecurity environments. Internships also encourage analytical thinking and provide exposure to solving security-related challenges using modern technologies..

Internships are beneficial for both students and organizations. While students gain practical knowledge, industry exposure, and professional experience, organizations are able to identify skilled and motivated individuals through project participation and teamwork. The internship experience also prepares students to adapt to workplace culture, deadlines, responsibilities, and collaborative environments, which are essential for future professional careers.

Outcomes:

- Upon successful completion of the internship, the intern will be able to:
- Apply theoretical and conceptual knowledge to practical Android application development projects.
- Demonstrate effective communication, teamwork, time management, and technical documentation skills.
- Gain practical exposure in web application security testing and vulnerability assessment.
- Understand the implementation of security scanning techniques such as SQL Injection and Cross-Site Scripting detection.
- Develop problem-solving abilities through innovative sustainability-focused applications.
- Document real-world cybersecurity use cases and develop solutions using modern technologies.

Organization: SuprMentr Technologies Private Limited

Nature of Work:

Industry-Oriented Cyber Security and Ethical Hacking Training with a focus on Web Application Security Testing, Vulnerability Assessment, Risk Analysis, and Secure Software Development.

Internship Project Title:

“Web App Security Testing Tool – Automate Testing for XSS and SQL Injection Vulnerabilities”

Description of Internship Work:

The project focused on the development of a web-based security testing application capable of identifying common web application vulnerabilities and security misconfigurations. The application was designed to perform automated vulnerability assessment through SQL Injection testing, Cross-Site Scripting detection, security header analysis, technology fingerprinting, port scanning, attack surface evaluation, and security risk assessment.

The major contributions during the internship included:

- Designing and developing a modern web-based cybersecurity dashboard using HTML, CSS, and JavaScript.
 - Developing the backend security assessment engine using Python and Flask.
 - Implementing SQL Injection testing techniques for identifying potential database-related vulnerabilities.
 - Implementing Cross-Site Scripting (XSS) detection mechanisms for analyzing input reflection behavior.
 - Developing security header analysis modules to evaluate HTTP security configurations.
 - Implementing technology fingerprinting techniques to identify web technologies used by target applications
 - Developing open port scanning functionality for identifying exposed services.
 - Creating risk scoring and attack surface analysis mechanisms for security evaluation.
 - Performing application testing, debugging, optimization, and UI enhancement activities.
-

Internship Objectives:

- i. To gain practical experience in Cyber Security and Ethical Hacking concepts and methodologies.
- ii. To understand and implement web application vulnerability assessment techniques.
- iii. To develop automated testing mechanisms for SQL Injection and Cross-Site Scripting vulnerabilities.
- iv. To design and implement security analysis modules such as header analysis, technology detection, and port scanning.
- v. To improve technical, professional, analytical, and problem-solving skills through real-world cybersecurity project development.
- vi. To acquire industry-oriented development practices, teamwork exposure, and project implementation experience for future professional growth in Cyber Security and Secure Software Development.

CHAPTER 3

ON JOB TRAINING

During the internship period at SuprMentr Technologies Private Limited, practical exposure was provided in the field of Cyber Security and Ethical Hacking, with a focus on web application security, vulnerability assessment, security testing methodologies, and secure software development practices. The internship mainly focused on developing the project titled “Web App Security Testing Tool – Automate Testing for XSS and SQL Injection Vulnerabilities” using modern web technologies and cybersecurity concepts.

The on-job training helped in understanding the complete lifecycle of web application development and security assessment, including project planning, user interface design, vulnerability analysis, risk assessment, testing, debugging, and deployment. Practical implementation of security header analysis, SQL Injection detection, Cross-Site Scripting (XSS) testing, technology fingerprinting, attack surface evaluation, and port scanning techniques was carried out during the internship period.

The training also enhanced problem-solving ability, technical communication, teamwork, analytical thinking, and professional work discipline through continuous project development activities and real-world cybersecurity implementation practices.

3.1 TOOLS AND TECHNOLOGIES

The following tools and technologies were used during the internship project development:

Visual Studio Code

Visual Studio Code was used as the primary Integrated Development Environment (IDE) for developing, debugging, testing, and maintaining the web application.

Python

Python programming language was used as the core backend development language due to its simplicity, flexibility, and extensive support for web development and cybersecurity applications.

HTML5

HTML5 was used for designing the structure and layout of the web application interface.

CSS3

CSS3 was used for creating responsive layouts, modern dashboard interfaces, and improving the overall user experience of the application.

Firebase Authentication

Firebase Authentication was used for implementing anonymous user authentication and secure cloud access.

JavaScript

JavaScript was used to implement dynamic functionalities, asynchronous communication, and real-time interaction between the frontend and backend.

Requests Library

The Requests library was used for sending HTTP requests and analyzing web application responses during vulnerability assessment.

BeautifulSoup

BeautifulSoup was used for parsing HTML content and extracting information required for security analysis and technology detection.

GitHub

GitHub was used for project version control, source code management, and project documentation.

3.2 LANGUAGES LEARNT

The internship helped in improving knowledge and practical implementation skills in the following programming and web technologies:

Python

Python was the primary programming language used for developing the security scanner, vulnerability assessment modules, backend processing, and risk analysis functionalities.

HTML

HTML was used for designing and structuring the web application's user interface and dashboard components.

CSS

CSS was used for styling the application, implementing responsive layouts, and creating a modern cybersecurity-themed dashboard.

SQL Concepts

Basic SQL concepts and database security principles were learned while understanding SQL Injection vulnerabilities and secure database interaction practices.

3.3 SKILLS ACQUIRED

The internship provided practical exposure and helped in acquiring various technical and professional skills.

Technical Skills Acquired

- Web Application Security Testing
 - Vulnerability Assessment
 - SQL Injection Detection
 - Cross-Site Scripting (XSS) Analysis
 - Security Header Analysis
 - Technology Fingerprinting
 - Open Port Scanning
 - Risk Assessment and Scoring
 - Flask Web Development
 - Frontend Development using HTML, CSS, and JavaScript
 - Debugging and Error Handling
-

Professional Skills Acquired

- Teamwork and Collaboration
- Communication Skills
- Problem Solving Ability
- Time Management
- Technical Presentation Skills
- Project Planning
- Professional Discipline
- Analytical Thinking
- Adaptability to New Technologies

The on-job training experience significantly improved both technical competency and professional confidence, helping in understanding industry-oriented cybersecurity practices, web application security assessment methodologies, and secure software development principles.

CHAPTER 4

USECASE IMPLEMENTATION

4.1 PROBLEM DEFINITION

Web applications play a critical role in today’s digital ecosystem, supporting key industries such as education, banking, healthcare, e-commerce, and government services. Their widespread use and internet accessibility make them highly attractive targets for cyber attackers aiming to exploit security weaknesses. As reliance on web applications continues to grow, ensuring their security has become increasingly important to protect sensitive data and maintain system integrity.

Common vulnerabilities, including Cross-Site Scripting (XSS), SQL Injection (SQLi), insecure configurations, missing security headers, and exposed network services, pose serious risks. These weaknesses can lead to unauthorized access, data breaches, information leakage, and full system compromise. However, many organizations—especially small businesses and educational institutions—often lack the necessary tools and expertise to identify and address these vulnerabilities early in the development lifecycle.

To overcome these challenges, the project “Web App Security Testing Tool – Automate Testing for XSS and SQL Injection Vulnerabilities” was developed. This tool offers an automated approach to security assessment by performing tasks such as SQL Injection testing, XSS detection, security header analysis, technology fingerprinting, port scanning, and attack surface evaluation. It provides a user-friendly, web-based platform that enhances security awareness, helps identify common vulnerabilities, and delivers actionable recommendations to strengthen the overall security posture of web applications.

4.2 ARCHITECTURE DIAGRAM

SYSTEM ARCHITECTURE DIAGRAM

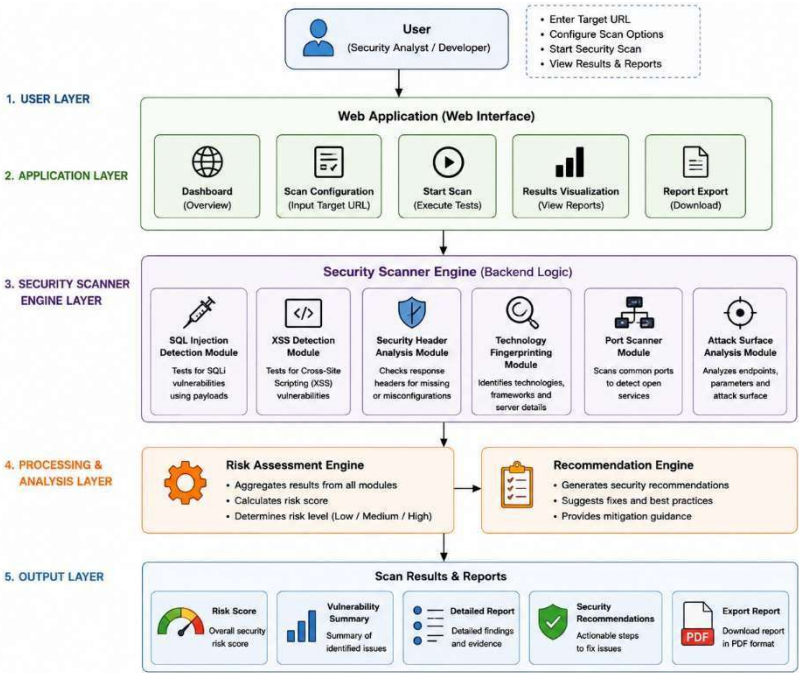


Figure 4.2.1

Components include in diagram:

User Layer

- User(Security Analyst/Developer)

Application Layer

- Displays scan statistics
- Shows recent scans and summaries

Security Scanner Layer

- SQL Injection Detection Module
- Cross Site Scripting(XSS) Detection Module
- Security Header Analysis Module
- Technology Fingerprinting Module
- Port Scanner Module
- Attack Surface Analysis Module

Processing & Analysis Layer

- Risk Assessment Engine
- Recommendation Engine

Output Layer

- Risk Score
 - Vulnerability Summary
 - Detailed Report
 - Security Recommendations
 - Export Report
-

4.3 SOFTWARE AND HARDWARE REQUIREMENTS

Table 4.3.1 : SOFTWARE REQUIREMENTS

Software	Purpose
Python 3.14	Core Programming Language
Flask	Backend Programming Language
HTML5	Frontend Structure
CSS3	User Interface Styling
JavaScript	Client-Side Functionality
Tailwind CSS	Responsive UI Design
MySQL	Database Management
Nmap	Port Scanning and Network Analysis
GitHub	Version Control

Table 4.3.2 : HARDWARE REQUIREMENTS

Hardware	Specification
Laptop/PC	Minimum 8GB RAM
Display	1366 x 768 Resolution or Higher
Internet Connection	Required
Processor	Intel i5 or above
Storage	Minimum 20GB Free Space

4.4 FUNCTIONAL AND NON FUNCTIONAL REQUIREMENTS

Table 4.4.1: FUNCTIONAL REQUIREMENTS

Requirement ID	Functional Requirement
FR1	Enter and Validate Target URL
FR2	Configure Security Scan Options
FR3	Perform SQL Injection Testing
FR4	Perform SQL Injection Testing
FR5	Analyze HTTP Security Headers
FR6	Perform Technology Fingerprinting
FR7	Conduct Port Scanning
FR8	Analyze Attack Surface
FR9	Generate Risk Assessment Score
FR10	Generate Security Recommendations
FR11	Display Scan Results Dashboard
FR12	Export Security Reports in PDF Format

Table 4.4.2 : NON-FUNCTIONAL REQUIREMENTS

Requirement ID	Non Functional Requirement
NFR1	User Friendly Interface
NFR2	Fast Scan Performance
NFR3	Scalability for Multiple Scans
NFR4	High Reliability
NFR5	Secure Data Processing
NFR6	Responsive Web Interface
NFR7	Report Generation Efficiency
NFR8	Cross-Browser Compatibility
NFR9	Availability and Accessibility
NFR10	Maintainability and Extensibility

4.5 USE CASE DIAGRAMS

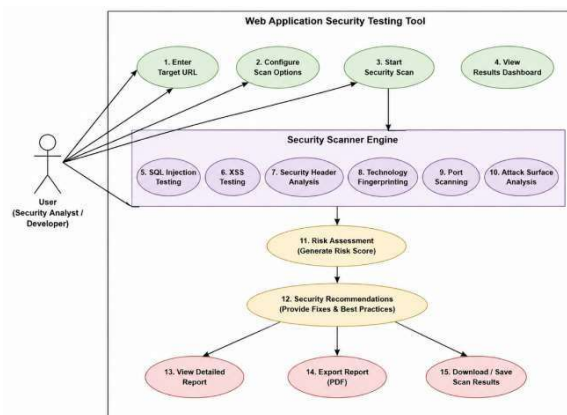


Figure: 4.5.1

4.6 DFD (DATA FLOW DIAGRAM)

DFD FLOW



Figure: 4.6.1

DFD COMPONENTS

External Entity

- User(Security Analyst / Developer)

Process

- Target URL Input
- Scan Configuration
- Security Scanning
- Vulnerability Analysis
- Risk Assessment
- Recommendation Generation
- Report Generation

Data Store

- Temporary Scan Results
- Generated Reports

Output

- Vulnerability Summary
- Detailed Security Report
- Risk Assessment Score
- Security Recommendations
- Exported PDF Report

CHAPTER 5

CODE SNIPPETS

5.1 Flask Application Initialization

```
1  from flask import Flask, render_template
2
3  app = Flask(__name__)
4
5  @app.route("/")
6  def home():
7      return render_template("index.html")
8
```

Description: This code initializes the Flask application and defines the home page route which loads the web dashboard.

Figure: 5.1.1

5.2 Website Scan API Endpoint

```
1  @app.route("/scan", methods=["POST"])
2  def scan():
3      data = request.get_json()
4      url = data.get("url")
5
6      results = scan_website(url)
7
8      return jsonify(results)
```

Description: This module receives the target URL from the user, invokes the security scanner and returns the scan results.

Figure 5.2.1

5.3 Security Header Analysis

```
1  headers_check = {
2      "Content-Security-Policy":
3          response.headers.get("Content-Security-Policy"),
4      "X-Frame-Options":
5          response.headers.get("X-Frame-Options"),
6      "Strict-Transport-Security":
7          response.headers.get("Strict-Transport-Security")
8  }
```

Description: This code checks whether important security headers are present in the target website response.

Figure 5.3.1

5.4 SQL Injection Detection Module

```
1 sql_payloads = [  
2     "' OR '1'='1",  
3     "' UNION SELECT NULL--",  
4     "' OR 1=1--"  
5 ]
```

Description: This module tests the target URL using common SQL Injection payloads and analyzes server responses.

Figure 5.4.1

5.5 Cross-Site Scripting (XSS) Detection

```
1 xss_payloads = [  
2     "<script>alert(1)</script>",  
3     "<img src=x onerror=alert(1)>",  
4     "<svg/onload=alert(1)>"  
5 ]
```

Description: This module uses predefined XSS payloads to identify possible reflected Cross-Site Scripting vulnerabilities.

Figure 5.5.1

5.6 Open Port Scanner

```
1 common_ports = [  
2     21, # FTP  
3     22, # SSH  
4     80, # HTTP  
5     443, # HTTPS  
6     8080 # Common alternate port  
7 ]
```

Description: This code checks commonly used network ports to identify exposed services on the target system.

Figure 5.6.1

5.7 Risk Assessment Engine

```
1  if risk_score >= 70:
2      risk_level = "HIGH"
3  elif risk_score >= 40:
4      risk_level = "MEDIUM"
5  else:
6      risk_level = "LOW"
```

Description: This module calculates the overall security risk score based on identified vulnerabilities and weaknesses.

Figure 5.7.1

5.8 Dashboard User Interface(HTML)

```
1  <div class="card">
2      <h3>Risk Score</h3>
3      <div id="riskScore" class="score-value">
4          0
5      </div>
6  </div>
```

Description: This code creates the dashboard section that displays the risk score of the scanned website.

Figure 5.8.1

5.9 JavaScript Scan Function

```
1  const response = await fetch("/scan", {
2      method: "POST",
3      headers: {
4          "Content-Type": "application/json"
5      },
6      body: JSON.stringify({ url: url })
7  });
```

Description: This function sends the target URL to the backend and updates the dashboard with the scan results.

Figure 5.9.1

5.10 Security Recommendations Module

```
1  recommendations.append(
2      "Add Content-Security-Policy header."
3  )
4  recommendations.append(
5      "Enable X-Frame-Options."
6  )
```

Description: This module generates security recommendations based on detected vulnerabilities and missing configurations.

Figure 5.10.1

CHAPTER 6

SCREENSHOTS

This chapter presents the important screenshots of the “Web App Security Testing Tool – Automate Testing for XSS and SQL Injection Vulnerabilities” developed during the internship period. The screenshots demonstrate the implementation of automated web application security assessment, vulnerability detection, security header analysis, technology fingerprinting, open port scanning, risk evaluation, and security recommendation generation functionalities. The system provides a professional dashboard interface for visualizing security metrics, attack surface analysis, detected vulnerabilities, security grades, and overall security posture of the target web application. These screenshots highlight the practical implementation of cybersecurity concepts, vulnerability assessment techniques, and secure software development practices used throughout the project.

6.1 Landing Page

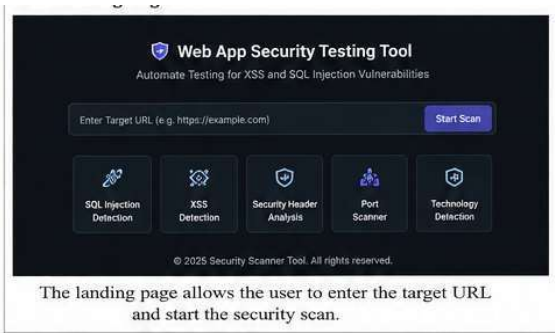


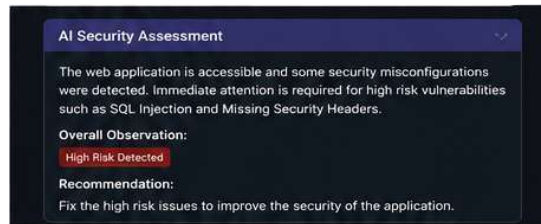
Figure 6.1.1

6.2 Security Scan Dashboard



Figure 6.2.1

6.3 AI Security Assessment



This section shows the AI generated security assessment and overall observation of the target website.

Figure: 6.3.1

6.4 Technologies Fingerprinting

Technologies Detected	
Technology	Details
Web Server	nginx/1.18.0
Programming Language	PHP 5.6.40
Framework	Bootstrap
JavaScript Libraries	jQuery 1.11.3
Analytics	Google Analytics
Other	Font Awesome

This module detects technologies, frameworks and components used in the target web application.

Figure: 6.4.1

6.5 Open Port Analysis

Open Port Analysis		
Port	Service	Status
21	FTP	Open
22	SSH	Open
80	HTTP	Open
443	HTTPS	Open
8080	HTTP-Alt	Closed

This section shows the open ports and services running on the target server.

Figure: 6.5.1

6.6 Security Header Analysis

Security Header Analysis		
Header	Status	Value
Content-Security-Policy	Missing	-
X-Frame-Options	Missing	-
X-Content-Type-Options	Present	nosniff
Strict-Transport-Security	Missing	-
Referrer-Policy	Present	no-referrer-when-downgrade

This module checks the presence of important HTTP security headers.

Figure: 6.6.1

6.7 Vulnerability Detection

Vulnerabilities Detected		
Vulnerability	Severity	Status
SQL Injection (GET)	High	Vulnerable
Cross Site Scripting (Reflected)	High	Vulnerable
Missing Security Headers	Medium	Detected
Server Information Disclosure	Low	Detected
Directory Listing Enabled	Low	Detected

This section lists the vulnerabilities found during the security scan.

Figure: 6.7.1

6.8 Security Recommendation

Security Recommendations	
✓	Use prepared statements to prevent SQL Injection.
✓	Validate and sanitize user input to prevent XSS attacks.
✓	Implement Content-Security-Policy header.
✓	Use X-Frame-Options header to prevent clickjacking.
✓	Enable HTTPS and Strict-Transport-Security header.

This section provides actionable recommendations to improve the security of the application.

Figure 6.8.1

6.9 Scan Logs

Scan Logs	
[10:30:15]	Starting scan for http://testphp.vulnweb.com
[10:30:16]	Checking target availability... OK
[10:30:18]	Detecting technologies... Completed
[10:30:25]	Analyzing security headers... Completed
[10:30:30]	Testing for SQL Injection... Completed
[10:31:05]	Testing for XSS vulnerabilities... Completed
[10:31:40]	Scanning open ports... Completed
[10:31:55]	Generating report... Completed
[10:32:00]	Scan completed successfully.

This section shows the detailed execution logs of the security scanner.

Figure: 6.9.1

6.10 Complete Security Assessment Report



This is the complete security assessment report generated after scanning the target website.

Figure: 6.10.1

CHAPTER 7

INTERNSHIP SUMMARY AND CONCLUSIONS

The internship at SuprMentr Technologies Private Limited provided valuable practical exposure in the field of Cyber Security and Ethical Hacking, web application security testing, vulnerability assessment, and secure software development practices. The internship mainly focused on the development of the project titled “Web App Security Testing Tool – Automate Testing for XSS and SQL Injection Vulnerabilities”, which was designed to identify common web application security weaknesses and provide automated security analysis through a user-friendly web-based platform.

During the internship period, practical knowledge was gained in Python programming, Flask web development, frontend development using HTML, CSS, and JavaScript, and cybersecurity assessment methodologies. The internship helped in understanding the complete software development lifecycle including project planning, system design, user interface development, security testing, debugging, performance optimization, and deployment. Practical implementation of security header analysis, SQL Injection detection, Cross-Site Scripting (XSS) testing, technology fingerprinting, attack surface evaluation, and open port scanning techniques provided valuable hands-on cybersecurity experience.

The project focused on automating web application security assessment by analyzing target websites and identifying potential vulnerabilities and security misconfigurations. Security header evaluation, vulnerability detection, risk assessment, technology identification, and recommendation generation modules were successfully developed and tested during the internship period. The project also improved understanding of web security principles, HTTP communication, network security concepts, and vulnerability assessment methodologies used in real-world cybersecurity environments.

The internship experience also provided exposure to practical cybersecurity challenges and security assessment approaches used to strengthen the security posture of web applications. The project demonstrated how modern web technologies and security testing techniques can be used to identify vulnerabilities, assess risks, and provide actionable recommendations for improving application security.

Overall, the internship was highly beneficial in strengthening technical competency, cybersecurity awareness, professional discipline, and practical industry exposure. The knowledge and experience gained during the internship will be valuable for future academic projects, cybersecurity research activities, ethical hacking practices, and professional career development in the fields of Cyber Security, Ethical Hacking, Secure Software Development, and Information Security.

Future enhancements to the project may include advanced vulnerability assessment techniques, integration with Artificial Intelligence for intelligent threat analysis, support for additional security testing modules, automated report generation, and enhanced penetration testing capabilities. These improvements can further increase the effectiveness of the system in identifying and mitigating web application security risks.

CONCLUSION

The internship carried out at **SuprMentr Technologies Private Limited** was a valuable learning experience that provided practical exposure to **Cyber Security, Ethical Hacking, Vulnerability Assessment, and Secure Web Application Development**. The successful implementation of the project “**Web App Security Testing Tool – Automate Testing for XSS and SQL Injection Vulnerabilities**” demonstrated the practical application of Python programming, Flask web development, web security testing methodologies, and vulnerability assessment techniques for identifying and mitigating common security risks in web applications..

The project successfully achieved its objective of developing an automated web application security assessment platform capable of performing security header analysis, SQL Injection testing, Cross-Site Scripting (XSS) detection, technology fingerprinting, open port analysis, risk evaluation, and security recommendation generation. The internship also enhanced practical understanding of software development methodologies, cybersecurity principles, HTTP communication, vulnerability assessment processes, secure coding practices, and modern web application architectures.

The overall internship experience contributed significantly towards technical skill enhancement, cybersecurity awareness, analytical thinking, problem-solving ability, professional growth, and industry-oriented learning. It provided valuable exposure to real-world security challenges and security assessment techniques, thereby creating a strong foundation for future career opportunities in **Cyber Security, Ethical Hacking, Secure Software Development, Information Security, and related emerging technology domains**.

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